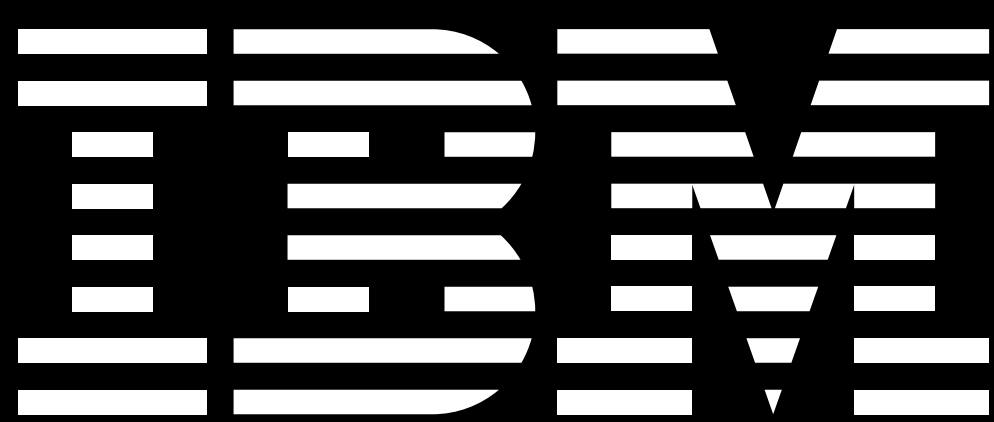


Context-Parallel Mamba2: Scaling to 1M+ Tokens

Garrett Goon



Mamba2: A Linear Attention Layer

The quadratic $\mathcal{O}(\text{seq\,len}^2)$ scaling of the standard Transformers attention mechanism has increasingly become a bottleneck as context lengths have ballooned with the advent of reasoning models and rising prevalence of modalities such as video and audio.

Mamba2 [1] is an alternative information propagation algorithm belonging to the steadily growing class of *linear* $\mathcal{O}(\text{seq\,len})$ attention mechanisms. Its GPU-aware design enables hardware utilization comparable to quadratic attention during training, and reduces the decoding time and cache-space from $\mathcal{O}(\text{seq\,len})$ to $\mathcal{O}(1)$. **GG: MODEL REFS! GG: Arch summary table**

Fully leveraging this improved scaling requires efficient training of Mamba2-based models on long-sequence documents, which poses engineering challenges due to the linear memory growth with context length. Context-parallelism, in which sequences are sharded along the sequence dimension across GPUs, is a natural and scalable approach to long-sequence training. This poster describes context-parallel implementation for Mamba2. **GG: Should probably mention ring attn somewhere**



Figure 1: Pink-necked green pigeon [2].

Mamba2 Architecture (Simplified)

Mamba2 relies on two central mechanisms for propagating information:

1. Short 1D causal convolutions **GG: github link**
2. A gated recursion relation

Letting $x_{sd} \in \mathbb{R}^{\text{seq\,len} \times \text{d}_{\text{model}}}$ be an input tensor, the Mamba2 outputs are of the schematic form

$$z_{sd} \sim \text{gated_recursion}(\text{causal_conv1d}(x_{sd}))$$

Both components, described in some more detail below, require adaptation in the context-parallel implementation. For brevity, we suppress batch and head dimensions throughout.

Causal Convolutions

The causal convolution is a depthwise, 1D convolution along the sequence dimension with a short filter, typically of width $K = 4$ [3]:

$$z_{sd} = \sum_{k=0}^K W_{dk} x_{(s-k)d} \cdot$$

Gated Recursion Relations

The central elements in the Mamba2 recursion relation are of the form:

$$z_{sd} = e^{-A_s} z_{(s-1)d} + \Delta_s x_{sd}$$

where the data-dependent A_s, Δ_s control the deletion and addition of information to the state z_{sd} . While the complete tensor z_{sd} can be constructed in $\mathcal{O}(\text{seq\,len})$ by solving the recursion in the naive manner, such an approach is suboptimal in practice as it cannot leverage GPU tensor cores. For this reason, the recursion is solved using a chunked, matmul-based strategy which has inferior theoretical, big- $\mathcal{O}$ scaling, but superior in-practice wall times [1].

Biological Information

Domain	Eukaryota
Kingdom	Animalia
Phylum	Chordata
Class	Aves
Clade	Columbimorphae
Order	Columbiformes
Family	Columbidae
Type genus	Columba

This box is styled differently compared to the others. To make such changes persistent across the whole poster, we can use these functions:

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#pop.update-poster-layout(. . .)
#pop.update-theme()
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Peace of Posters Documentation

You can find more information on the documentation site under jonaspleyer.github.io/peace-of-posters/.

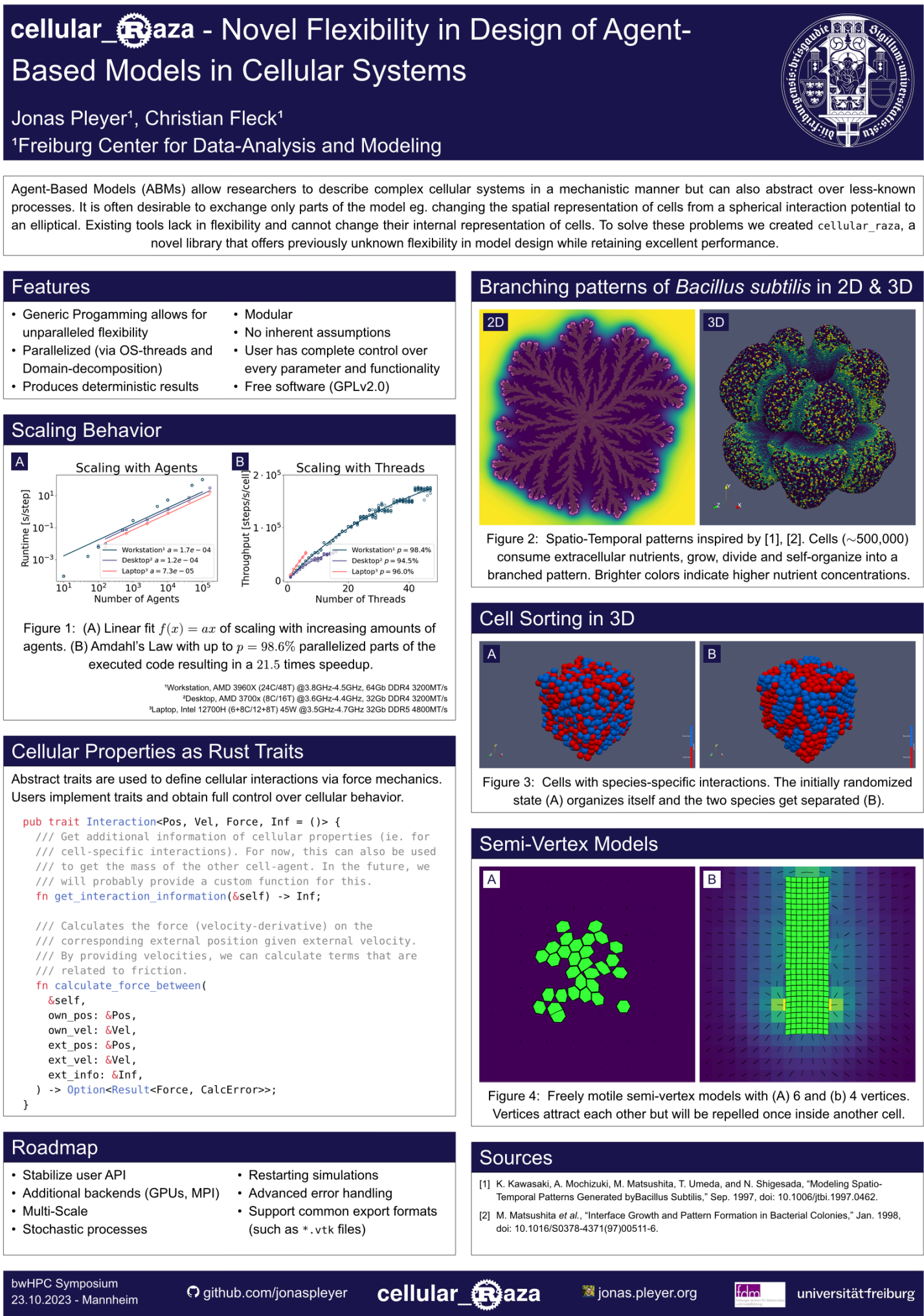


Figure 2: The poster from the thumbnail can be viewed at the documentation website as well.

General Relativity

Einstein’s brilliant theory of general relativity starts with the field equations [4].

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa T_{\mu\nu}$$

However, they have nothing to do with doves.

Peace be with you



Figure 3: ‘Doves [...] are used in many settings as symbols of peace, freedom or love. Doves appear in the symbolism of Judaism, Christianity, Islam and paganism, and of both military and pacifist groups.’ [5].

Etymology

Pigeon is a French word that derives from the Latin pīpiō, for a ‘peeping’ chick, while dove is an ultimately Germanic word, possibly referring to the bird’s diving flight. The English dialectal word culver appears to derive from Latin columba [6]. A group of doves is called a “dule”, taken from the French word deuil (‘mourning’) [7].

References

[1] T. Dao and A. Gu, “Transformers are SSMS: Generalized Models and Efficient Algorithms Through Structured State Space Duality.” [Online]. Available: <https://arxiv.org/abs/2405.21060>

[2] Wikipedia, “File:Treron vernans male - Kent Ridge Park.jpg — Wikipedia, The Free Encyclopedia.” 2024.

[3] “causal-conv1d.” [Online]. Available: <https://github.com/Dao-AILab/causal-conv1d>

[4] A. Einstein, “Die Grundlage der allgemeinen Relativitätstheorie [AdP 49, 769 (1916)],” *Annalen der Physik*, vol. 517, no. S1, pp. 517–571, 1916, doi: 10.1002/andp.2005517s151.

[5] Wikipedia, “Doves as symbols — Wikipedia, The Free Encyclopedia.” 2024.

[6] Wikipedia, “Online Etymology Dictionary — Wikipedia, The Free Encyclopedia.” 2024.

[7] J. Lipton, *An Lipton James : Exaltation of larks*. Harlow, England: Penguin Books, 1991.

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Notice that this box would not fill the entire space up to the bottom of the page but we can stretch it such that it does so anyway.